

Dataset Overview

Project Introduction

This project is based on “**Logistics and Supply Chain domain**”, mainly focusing on “**Inventory Management**”.

The main goal of this project is to analyze warehouse data and understand how inventory operations are working. I tried to find problems like inefficiencies, mismatches in inventory, and process issues, and then derive useful insights from the data.

In real-world supply chain systems, inventory is very important because if it is not managed properly, it can lead to problems like stockouts, overstocking, higher costs, and delays.

So, through this project, I focused on analyzing different operations like:

- Inventory movement
- Discrepancies
- Validation process
- Task allocation across warehouses

Data Source

The dataset used in this project is a **simulated dataset** which represents real warehouse operations.

- Data Type: CSV (multiple sheets)
- Time Period: Jan 2021 to Dec 2024

There are **7 different tables**, and each table represents a different part of warehouse operations.

Tables Description

1. **Inventory Discrepancy Log -**
This table shows where the expected inventory and actual inventory do not match. It helps identify errors or issues.
2. **Inventory Item Master -**
This is a master table which contains basic details of items like category and priority.
3. **Inventory KPI Summary -**
This table shows overall performance of warehouses like total movements, discrepancies, etc.
4. **Inventory Movement Log -**
This table shows how items are moving inside the warehouse, from one location to another. It helps us understand the flow of inventory.
5. **Inventory Profiling Dataset -**
This table categorizes items based on their movement and demand patterns.
6. **Inventory Validation Log -**
This table shows whether inventory movements were properly checked or not (pass/fail).
7. **Task Allocation Log -**
This table contains tasks given to staffs like picking, packing, etc. It helps analyze how efficiently tasks are completed.

Column Description

The dataset contains different types of columns:

- ❖ **IDs (Identifiers)** :- item_id, movement_id, task_id, profiling id, etc.
- ❖ **Categorical** :- warehouse, task type, discrepancy type, etc.
- ❖ **Numerical** :- quantity, variance, etc.
- ❖ **Date/Time Variables** :- movement date, task date, etc.

Each column plays a significant role in understanding the dataset.

For example, movement_quantity helps us to understand how much inventory is moving while discrepancy_type helps us to understand what kind of error occurred.

Data Relationships

All tables are connected with each other using common columns like:

- item_id, connects item-related tables
- movement_id, connects movement, discrepancy, and validation
- warehouse_code, connects warehouse-level data

Because of this, we can combine multiple tables and do deeper analysis.

Initial Observations

When I saw the dataset first time, I found:

- All columns, do not have any null values (All columns have full data)
- Presence of missing values in certain columns like validation remarks
- Duplicate entries in task allocation logs
- Some warehouses have more discrepancies than others
- There are some extreme values in movement quantity (outliers)

So before doing any analysis, data cleaning was required by basic norms.